Standardization in Machine Learning

Definition:
Standardization in Machine Learning is a data preprocessing technique used to transform features so that they have a mean of 0 and a standard deviation of 1. It helps machine learning algorithms perform better by bringing all features to the same scale.

Formula:

$$z = (x - \mu) / \sigma$$

Where:
• x = original value
• μ = mean of the feature
• σ = standard deviation
• z = standardized value

Why Standardization is Important:
• Improves performance of machine learning algorithms.
• Prevents features with large values from dominating the model.
• Useful for algorithms like KNN, SVM, Logistic Regression, Neural Networks, and PCA.

Difference Between Standardization and Normalization:
• Standardization: Mean = 0 and Standard Deviation = 1.
• Normalization: Scales values usually between 0 and 1.

Python Example using scikit-learn:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```